VICTORY 2027

# The Nervous System

LECTURE- 05

BIOLOGY

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# Topics to be Covered

1. Transmission of the Nerve Impulse
2. Types of Neurons


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

# Midbrain

- A small tubular part (reflexes involving eyes and ears).
- The midbrain controls reflex movements of the head, neck and trunk in response to visual and auditory stimuli.
- It also controls the reflex movements of the eye muscles, changes in pupil size and shape of the eye lens.

# Peripheral Nervous System (PNS)

## Somatic Nervous System

- Cranial nerves (12 pairs)
- Spinal nerves (31 pairs)

## Autonomic Nervous System (ANS)

- Sympathetic (nerves between neck and waist) **(Stimulatory)**
  Accelerates heart beta, dilates, pupil, constricts blood vessels, releases sugar from liver
- Autonomic Nervous System (ANS) (Nerves above neck and below sacrum) **(Calming)**
  Restores normal body conditions

# Emotions and Autonomic Nervous System

- Autonomic nervous system is strongly influenced by emotions such as grief, anger, fear, sexual stimulation, etc.
- High blood pressure, stomach ulcers and some other troubles may arise due to long continued emotional stress.

# Peripheral Nervous System (PNS)

**Peripheral nervous system (PNS)** includes the nerves that emerge from and enter into the brain and spinal cord.

A. **Somatic nervous system (SNS)** conveys information to skeletal (voluntary) muscles.

**Cranial nerves:**

- Emerge from the brain.
- 12 pairs of cranial nerves.
- **Sensory -** olfactory, optic and auditory.
- **Motor -** going to eye muscle.
- **Mixed -** going to and coming from face and tongue.

**Spinal nerves**

- Emerge from the spinal cord 31 pairs of spinal nerves
- Neck(8), thorax (12), lumbar region (5), sacral (5), coccygeal region (1)

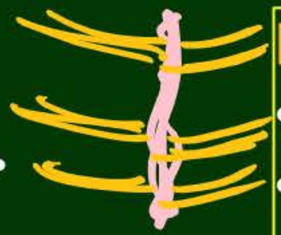
# Peripheral Nervous System (PNS)

- **Peripheral nervous system (PNS)** includes the nerves that emerge from and enter into the brain and spinal cord.

A. Autonomic nervous system (ANS).

B. Consists of a pair of chains of nerves and ganglia on either side of the backbone.

C. Controls involuntary actions.

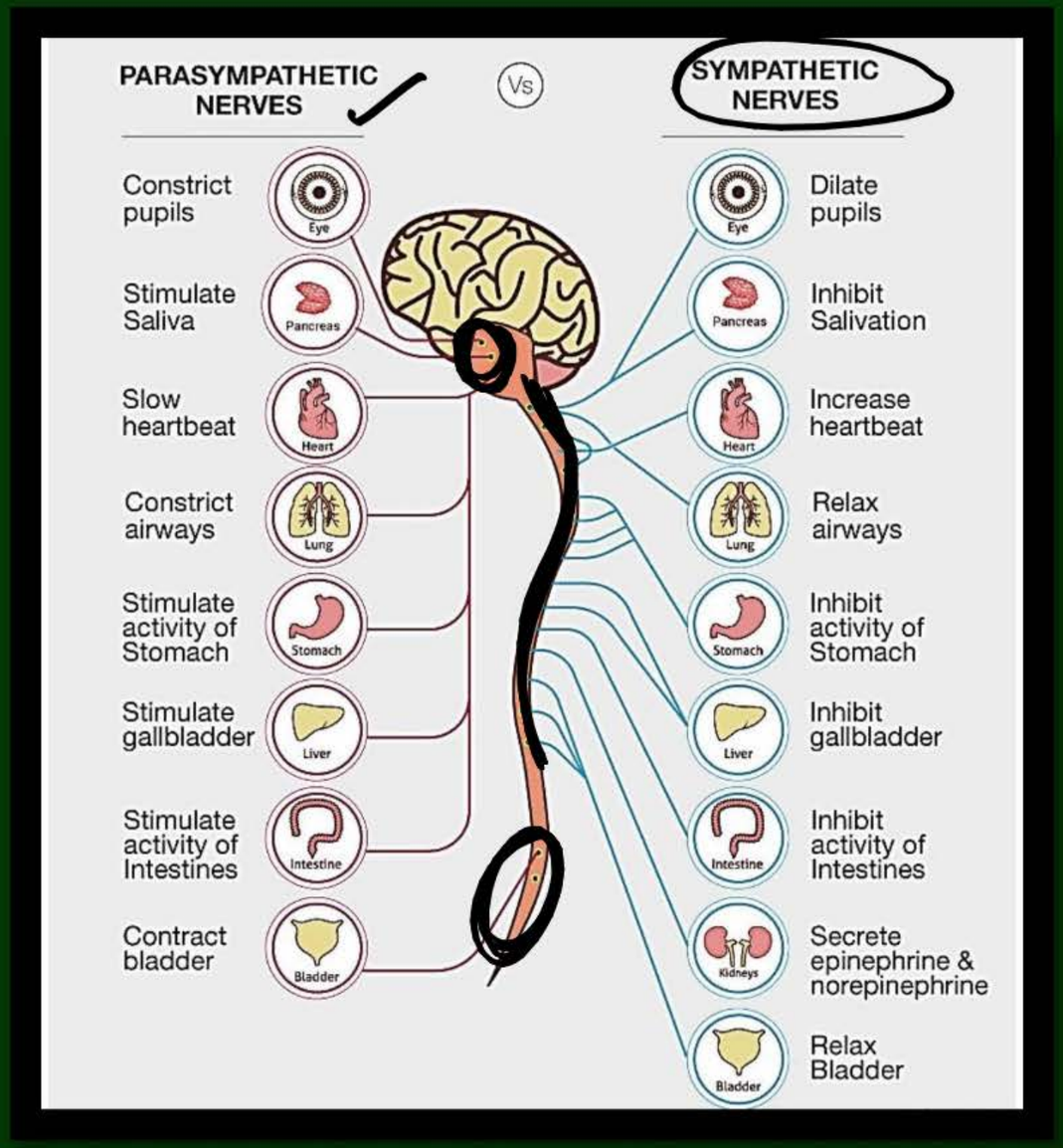


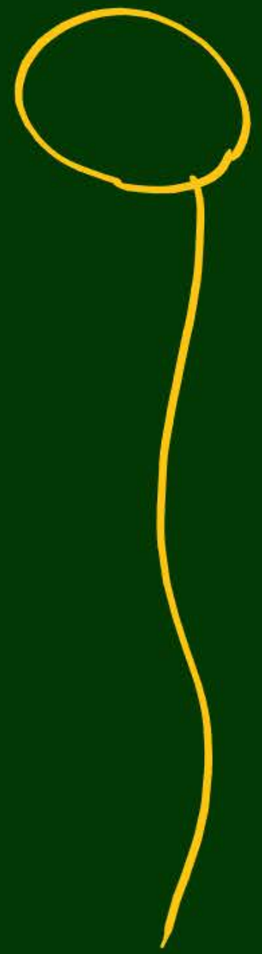
**Parasympathetic nervous sytem**

- It is located at 2 places
- Anteriorly in head and neck
- Posteriorly in sacral region

**Sympathetic nervous system:**

- Nerves arise from the spinal cord between Neck and waist region Stimulated by adrenaline hormone.


PARASYMPATHETIC NERVES
Vs
SYMPATHETIC NERVES
Constrict pupils
Eye
Dilate pupils
Stimulate Saliva
Pancreas
Inhibit Salivation
Slow heartbeat
Heart
Increase heartbeat
Constrict airways
Lung
Relax airways
Stimulate activity of Stomach
Stomach
Inhibit activity of Stomach
Stimulate gallbladder
Liver
Inhibit gallbladder
Stimulate activity of Intestines
Intestine
Inhibit activity of Intestines
Contract bladder
Bladder
Kidneys
Secrete epinephrine & norepinephrine
Relax Bladder



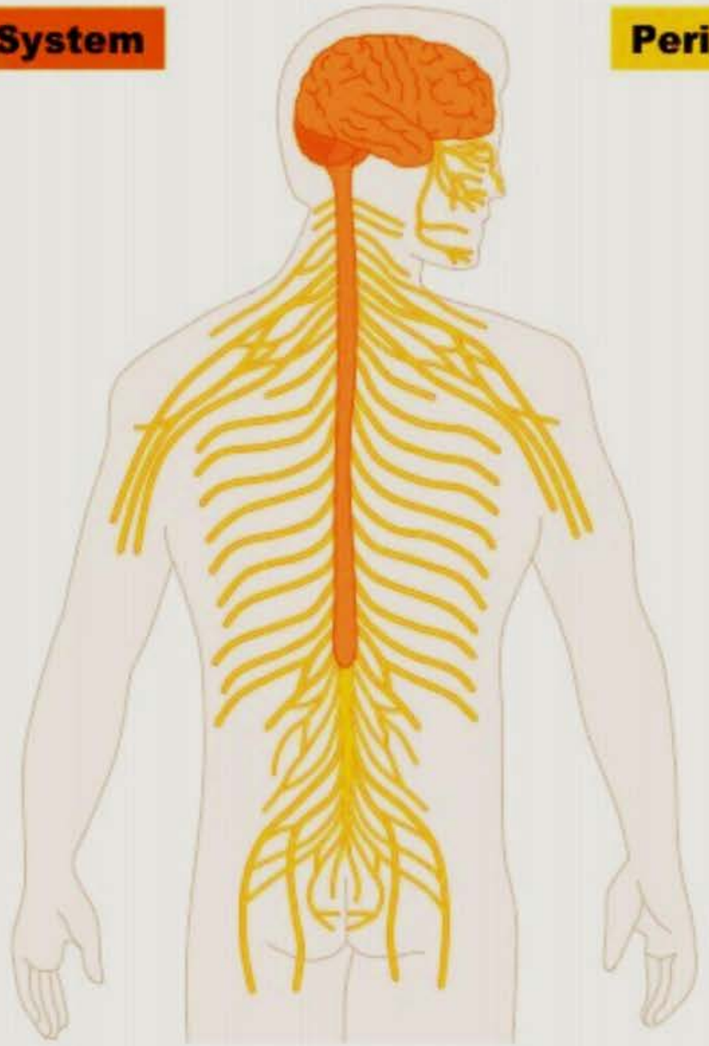
## Central Nervous System

**Composed of:**

- Brain
- Spinal cord

**Contains:**

- Relay neurons (interneurons)



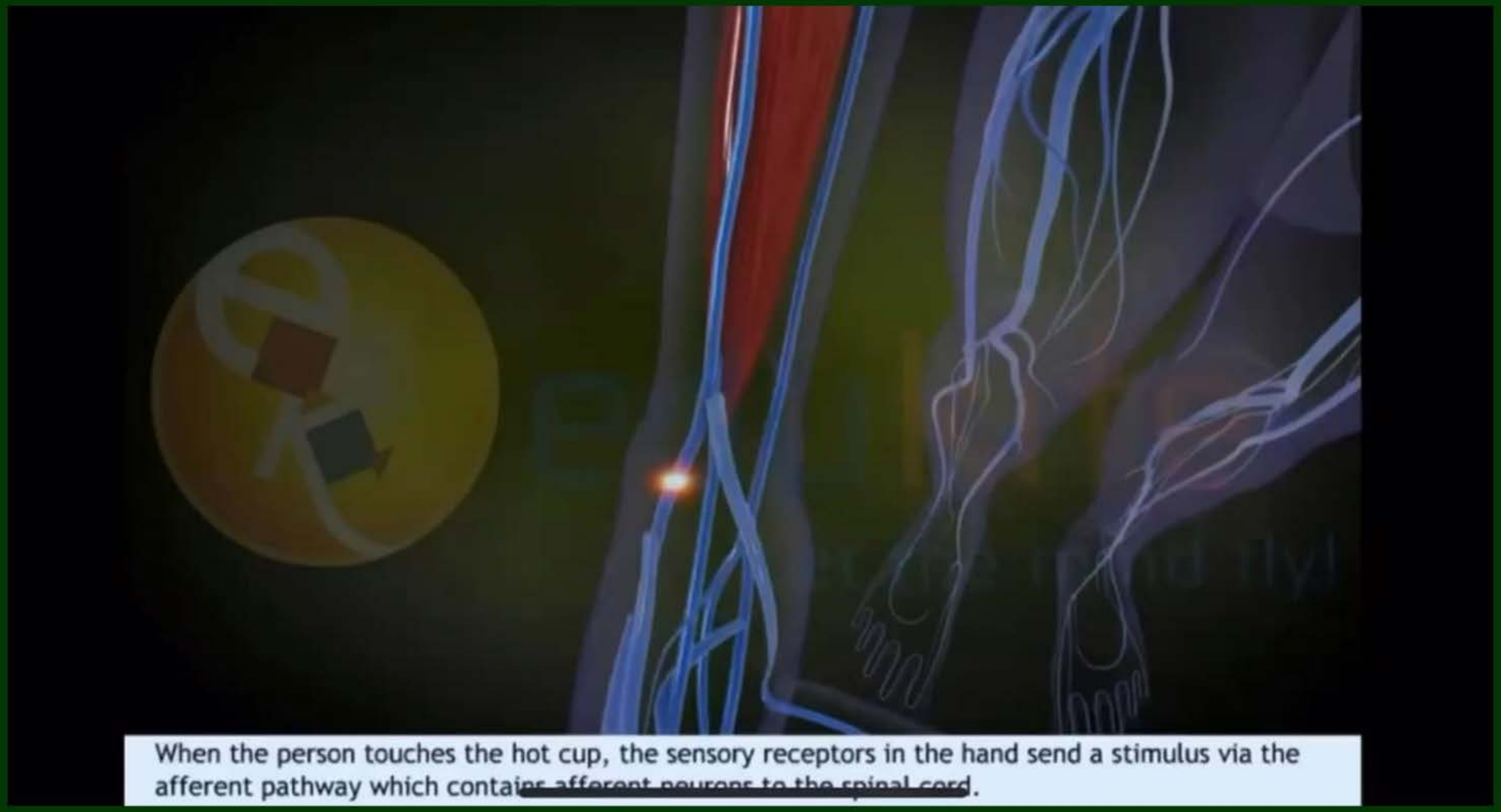
## Peripheral Nervous System

**Composed of:**

- Cranial nerves
- Spinal nerves
- Peripheral nerves

**Contains:**

- Sensory neurons
- Motor neurons


When the person touches the hot cup, the sensory receptors in the hand send a stimulus via the
afferent pathway which contains afferent neurons to the spinal cord.

## REFLEX

- The term "reflex" comes from the Latin word 'reflexus' meaning reflected or directed back.
- There are two types of actions which occur in our body:
- Voluntary (performed consciously) and
- Involuntary (performed unconsciously).

**Examples:**

**1. Voluntary actions**

- To watch some programme on TV.
- Pick up an apple and eat it.
- Know the time, you raise your arm.

**2. Involuntary actions** (can be slow & fast)

- It is an automatic/quick/immediate, involuntary action in the body brought about by a stimulus.
- Some particle falls into your eye and there is immediate flushing of tears to wash out the particle (glandular secretion).
- Instantaneous withdrawal of hand when it accidentally touches a hot iron (muscular movement). (Reflex action)

**A reflex arc may be represented as follows :**

❖ Stimulus → receptor in the sense organ → afferent (sensory) nerve fibre → CNS (spinal cord/brain) → efferent (motor) nerve fibre → muscle (to contract) / gland (to secrete).)

## Types of reflexes:

1. **Natural (inborn) reflex:**
   - Is one in which no previous experience or learning is required.
   - These reflexes are inborn, i.e., inherited from the parents.

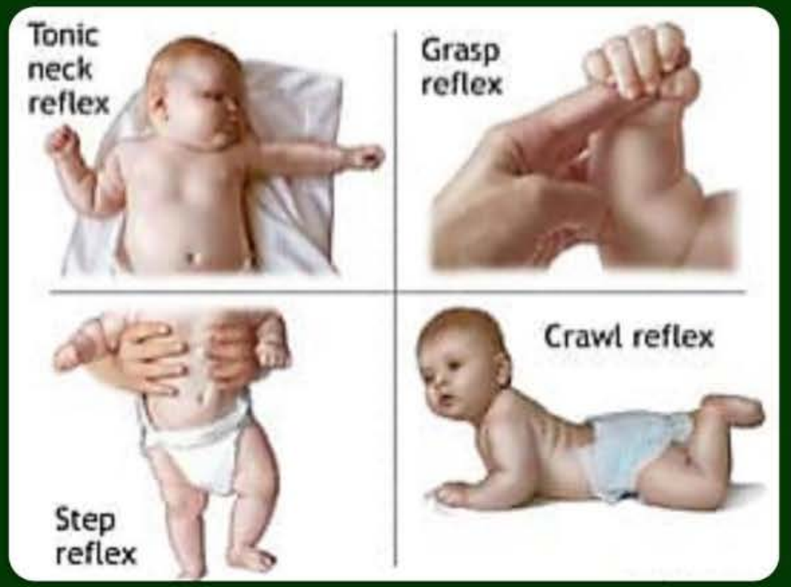
**Examples :**

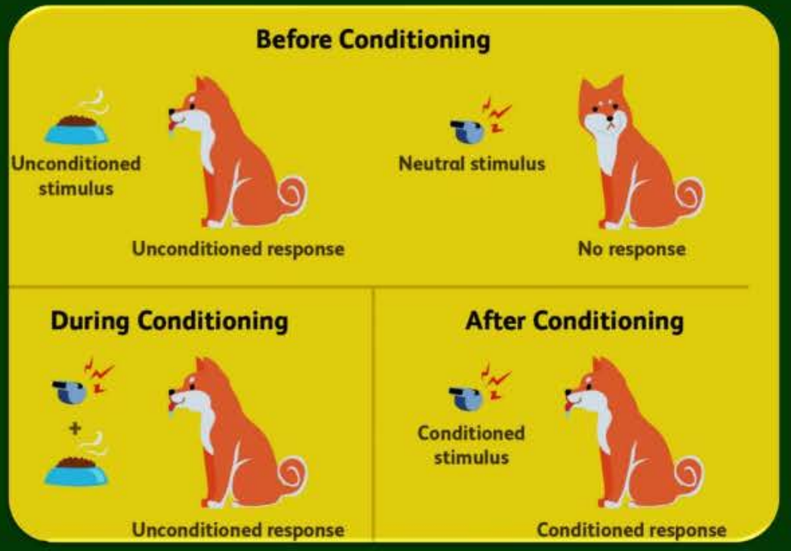
- Blinking, coughing, sneezing ; peristalsis
- These are protective.
- Salivation, swallowing, peristalsis : These provide functional efficiency.

2. **Conditioned (or acquired) reflex:**
   - Is one which develops during lifetime due to experience or learning.
   - For example, what you experience simply at the sight or by the smell of a familiar tasty food resulting in watering (salivation) of your mouth.

'Pavlov'

'Acquired'



Definitions

- **Response:** The change in an organism resulting due to stimulus.
- **Impulse:** A wave of irritability (electrical disturbance that sweeps over the nerve cell).
- **Receptors:** the specialised epithelial cells which, on receiving the stimulus, set up waves of impulses towards the central nervous system. e.g. photoreceptors (sight), phonoreceptors (sound), thermoreceptors (temperature), olfactory receptors (smell) and so on present in our sense organs.
- **Effectors:** muscles or glands which, on receiving the impulse from the brain or spinal cord, contract or secrete substances.

## QUESTION

Corpus callosum connects

A. Two cerebral hemispheres

B. Diencephalon and mid brain

C. Pons and medulla

D. Medulla and spinal cord

## QUESTION

The inner covering of human brain is ?

A Coracoid

B Piamater

C Duramater

D Arachnoid

## QUESTION

The outer covering of human brain is ?

A Coracoid

B Piamater

C Duramater

D Arachnoid

## QUESTION

Part of the brain that helps in regulation and maintenance of muscular activities

A Medulla oblongata

B Midbrain

C Cerebrum

D Cerebellum

## QUESTION

The Myelin sheath is derived from the

A Microglia

B Neuroglial cells

C Schwann cells

D Nerve cells

## QUESTION

_____ are the neurons carrying impulses away from the central nervous system.

A Efferent nerves

B Afferent nerves

C Extensors

D Sensory nerves

## QUESTION

Which of these has the highest permeability in a resting nerve cell?

A $Cl^-$

B $Na^+$

C $K^+$

D $I^-$

## QUESTION

A phase of cell division where the spindle fibres disappear is:

A Telophase

B Prophase

C Metaphase

D Anaphase

# QUESTION

Assertion (A): Arachnoid layer helps in speeding up the transmission of nerve impulses along the axons.

Reason (R): Axon endings at the synapse secrete acetylene.

A (A) is true and (R) is false.

B (A) is false and (R) is true.

C Both (A) and (R) are true.

D Both (A) and (R) are false.

# QUESTION

Many nuclear power plants are constructed near water sources. They use water for cooling their machinery and release it back into the water stream. Identify the kind of pollution caused.

A Radiation pollution

B Water pollution

C Industrial pollution

D Thermal pollution

# QUESTION

Vincet's day begins with a cup of piping hot coffee. Caffeine is a stimulant present in coffee which increases the circulation of Adrenaline in the body. Vineet feels refreshed, alert and focussed. [3]

(a) Name the organ that is stimulated by caffeine which makes Vineet alertand focussed.

(b) Give the collective term for the protective membranes of the organ mentioned in (a)

(c) Which part of the autonomic nervous system increases the secretion of Adrenaline? SNS



# QUESTION

Riya accidentally touches a hot pan and immediately pulls her hand away.

a. Name the type of response.

b. Which part of the nervous system coordinates it?

c. Why is it faster than a voluntary action?

(A) Voluntary
(B) Involuntary
(C) Reflex

Reflex → Spinal Cord

(A) BRAIN

(B) SPINAL CORD

# QUESTION

A doctor taps below a patient's kneecap, causing the lower leg to move forward automatically.

a. What type of action is this?

b. Is it voluntary or involuntary?

c. Name the pathway followed by the nerve impulse.

Reflex

A B

Reflex arc.

# QUESTION

A person steps on a sharp object and quickly withdraws the foot. The person becomes consciously aware of the pain shortly afterwards.

a. Why does withdrawal occur before conscious awareness of pain?

b. Which organ coordinates the reflex?

c. Give one advantage of this response.

# Thank You